

GateGuard: QR-Based Entry and Exit Management System

CS 432 – Databases
Indian Institute of Technology Gandhinagar
Semester II (2025–2026)

Team Almonds!

[Link to the GitHub Repository](#)

Sr. No.	Name	Roll Number
1	Harsh Jamgaonkar	23110130
2	Kavya Lavti	23110164
3	Kushal Mehta	23110205
4	Nilay Verma	23110219
5	Soham Gaonkar	23110314

Contents

1	Project Scope	3
1.1	Objective	3
1.2	In-Scope Features	3
2	Core Functionalities	3
2.1	QR Code Management	3
2.2	Entry and Exit Logging	4
2.3	Vehicle Management	4
2.4	Guest Management	4
2.5	Visitor Management	4
2.6	Gate and Guard Management	4
2.7	Blacklist Management	5
3	Domain Constraints and Business Rules	5
3.1	Identity Constraints	5
3.2	Entry and Exit Rules	5
3.3	Presence Tracking Rule	5
3.4	Guest Constraints	5
3.5	Vehicle Constraints	6
4	Database Schema Overview	6
5	Conclusion	7

1 Project Scope

1.1 Objective

The objective of the **GateGuard System** is to design and implement a QR-based Entry and Exit Management System for IIT Gandhinagar. The system aims to:

- Digitally manage entry and exit of members (residents), visitors (Amazon, BigBasket, cab drivers), guests, and non-resident staff.
- Track vehicles entering and exiting the premises.
- Prevent unauthorised access.
- Maintain detailed logs for each gate.
- Generate daily and weekly traffic reports.

1.2 In-Scope Features

The system includes the following functionalities:

- Member registration (students, faculty, staff, guests).
- QR code generation and assignment.
- Entry and exit logging using QR scan.
- Vehicle registration and vehicle-specific QR assignment.
- Guest pre-registration linked to host members.
- Blacklist management and violation control.
- Real-time campus presence tracking.
- Gate-wise logging of activities.
- QR scan displays associated person image from database.
- Expired QR detection with alert mechanism to guard.

2 Core Functionalities

2.1 QR Code Management

- Generate unique QR for each person (member, guest, visitor, staff).
- Define validity period (Valid_Till).
- Record issue date.
- Validate QR during scan.
- Reject expired QR codes.

2.2 Entry and Exit Logging

- Scan QR at the gate.
- Verify identity and vehicle details.
- Check blacklist status.
- Log entry/exit with timestamp.
- Update real-time campus presence status.
- Prevent duplicate entry without exit.

2.3 Vehicle Management

- Register vehicles linked to members.
- Assign vehicle-specific QR codes.
- Maintain vehicle number and type.
- Track vehicle campus presence.
- Log vehicle entry and exit separately.

2.4 Guest Management

- Pre-registration of guests.
- Link guest to host Member_ID.
- Record visit reason and details.
- Approval/rejection by authorities.
- Time-bound guest validity.
- Maintain guest entry and exit logs.

2.5 Visitor Management

- Temporary visitor registration at arrival.
- Collect personal and vehicle details.
- Generate QR valid for 6 hours.

2.6 Gate and Guard Management

- Each gate has unique Gate_ID.
- Maintain opening and closing times.
- Each guard has unique Guard_ID.
- Guard shift timings are recorded.

2.7 Blacklist Management

- Add violating members to blacklist.
- Define blacklist expiry.
- Automatically deny entry if blacklist active.

3 Domain Constraints and Business Rules

3.1 Identity Constraints

- Each member has a unique Member_ID.
- Each QR code belongs to exactly one person.
- A member may own multiple vehicles.
- Each vehicle belongs to exactly one member.
- Each vehicle has a unique Vehicle_ID.

3.2 Entry and Exit Rules

- A person cannot enter twice without exiting.
- A person cannot exit without first entering.
- QR validity condition: Current Date $\leq$ Valid_Till.
- Entry denied if blacklist expiry date $>$ current date.
- Gate must exist before log creation.
- Entry must occur within gate operating hours.

3.3 Presence Tracking Rule

The system maintains an `inCampusFlag`:

- $1 \rightarrow$ Currently inside campus.
- $0 \rightarrow$ Outside campus.

The flag updates atomically during entry and exit operations.

3.4 Guest Constraints

- Each guest must be linked to a host member.
- Guest access must be approved.
- Guests cannot enter without approval.

3.5 Vehicle Constraints

- Vehicle QR must be valid during scan.
- Vehicle must belong to a registered member.
- Vehicle entry must be logged with Gate_ID and timestamp.
- Vehicle cannot enter twice without exit.

4 Database Schema Overview

This section presents the database tables along with their Primary Keys (PK) and Foreign Keys (FK). Referential integrity is enforced through foreign key constraints, and all tables include at least three NOT NULL attributes to maintain data consistency.

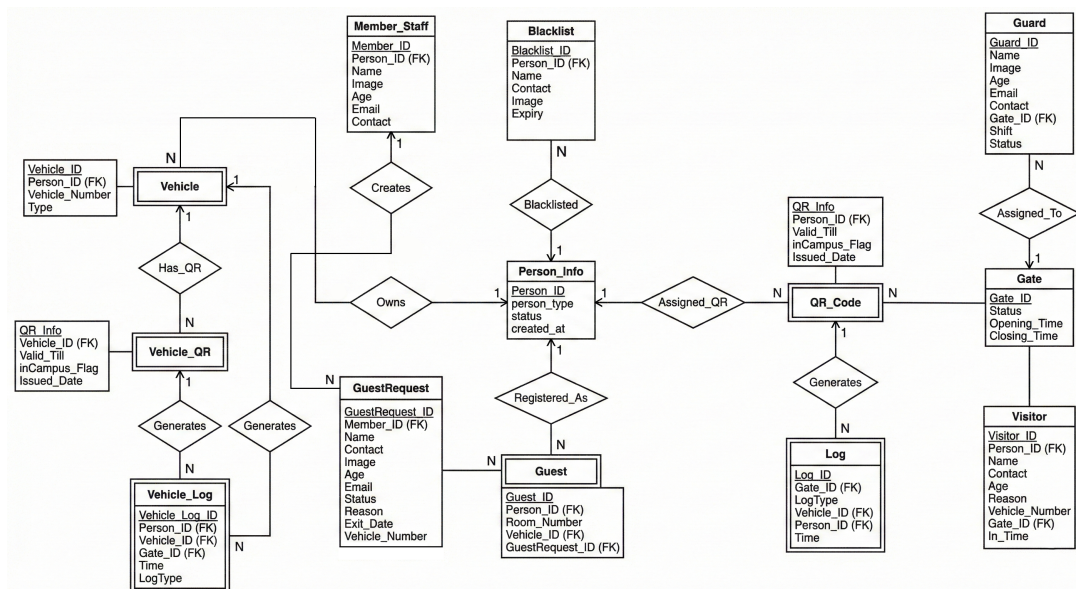


Figure 1: Entity-Relationship (ER) Diagram of the GateGuard System



Figure 2: Database Tables with Primary and Foreign Keys

5 Conclusion

The GateGuard system provides a scalable and secure QR-based campus access management solution. The design ensures data integrity, real-time tracking, controlled access, and efficient monitoring of campus movement.